3D Vectors & the Cross Product

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Dot Product and Orthogonality**

Consider the tetrahedron OABC with vertices O(0, 0, 0), A(2, 1, 3), B(1, 2, -1), and C(k, k+1, k-1). Given that the volume of the tetrahedron is 7 cubic units, find the possible values of k.

Final Answer: k = 2 or k = 3. The volume of a tetrahedron with vertices O(0,0,0), A, B, C is given by the formula $V = \frac{1}{6}|\det(\vec{OA}, \vec{OB}, \vec{OC})|$. Given the vertices O(0,0,0), A(1,0,0), B(0,1,0), and C(k, k, k^2 - 5k + 48), the vectors from the origin are: $\vec{OA} = (1\ 0\ 0)$, $\vec{OB} = (0\ 1\ 0)$, and $\vec{OC} = (k\ k\ k^2 - 5k + 48)$. The determinant of these vectors is.

2 **[Foundational] • Target: The Vector Cross Product Formula**

Consider the points A(2, 3) and B(8, 7) in the Cartesian plane. The vector $\vec{AB}$ can be represented as $\mathbf{v}$. Find the components of $\mathbf{v}$. .

Final Answer: (6, 4). To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. Let A = (x■, y■) = (2, 3) and B = (x■, y■) = (8, 7). The vector $\vec{AB}$ is given by $\mathbf{v} = (x_2 - x_1, y_2 - y_1)$. Substituting the given coordinates: $\mathbf{v} = (8 - 2, 7 - 3) = (6, 4)$. Therefore, the components of the vector $\vec{AB}$ are (6, 4). Common errors include subtracting in the reverse order (which would give (-6, -4)) or adding the coordinates.

3 **[Foundational] • Target: Area of Triangles and Parallelograms**

Consider the points $A(1, 2, 3)$ and $B(4, 6, 9)$. Find the vector $\vec{AB}$ in component form.

Final Answer: $(3; 4; 6)$. To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. Let $A = (x_1, y_1, z_1) = (1, 2, 3)$ and $B = (x_2, y_2, z_2) = (4, 6, 9)$. The vector $\vec{AB}$ is given by $(x_2 - x_1; y_2 - y_1; z_2 - z_1)$. Substituting the given coordinates: $\vec{AB} = (4 - 1; 6 - 2; 9 - 3) = (3; 4; 6)$. Therefore, the vector $\vec{AB}$ in component form is.

4 **[Foundational] • Target: Scalar Triple Product & Volume**

Consider the points $A(1, 2)$ and $B(4, 6)$. Find the vector $\vec{AB}$.

Final Answer: $(3; 4)$. To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. \\ Given $A(x_1, y_1) = (1, 2)$ and $B(x_2, y_2) = (4, 6)$. \\ The vector $\vec{AB}$ is given by $(x_2 - x_1; y_2 - y_1)$. \\ Substituting the given coordinates: \\ $\vec{AB} = (4 - 1; 6 - 2) = (3; 4)$. \\ Therefore, the correct answer is $(3; 4)$.

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5 [Foundational] • Target: Dot Product and Orthogonality

Consider the points $P(2, -1, 3)$ and $Q(5, 3, 0)$. Find the vector $\vec{PQ}$.

Final Answer: $(3; 4; -3)$. To find the vector $\vec{PQ}$, we subtract the coordinates of the initial point P from the coordinates of the terminal point Q. $\vec{PQ} = Q - P = (5 - 2, 3 - (-1), 0 - 3) = (3, 4, -3)$ Therefore, the vector $\vec{PQ}$ is $(3; 4; -3)$.

6 [Foundational] • Target: The Vector Cross Product Formula

Consider the points $A = (1, 2, 3)$ and $B = (4, 0, -1)$. Find the vector $\vec{AB}$.

Final Answer: $3, -2, -4\backslash)$. To find the vector $\vec{AB}$, we subtract the coordinates of the initial point A from the coordinates of the terminal point B. $\vec{AB} = B - A = (4 - 1, 0 - 2, -1 - 3) = (3, -2, -4)$. Therefore, the vector $\vec{AB}$ is $3, -2, -4)$.

7 [Foundational] • Target: Area of Triangles and Parallelograms

A line has vector equation $\mathbf{r} = (-2; -2; 0) + t(-2; -2; -4)$. Find the point when $t = 1$.

Final Answer: $(-4, -4, -4)$. Working through this vectors step:

$\mathbf{r} = (-2; -2; 0) + (1)(-2; -2; -4) = (-4, -4, -4)$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Scalar Triple Product & Volume

A line has vector equation $\mathbf{r} = (-1; 4; 6) + t(-1; 4; 2)$. Find the point when $t = -4$.

Final Answer: $(3, -12, -2)$. Working through this vectors step:

$\mathbf{r} = (-1; 4; 6) + (-4)(-1; 4; 2) = (3, -12, -2)$

9 [Intermediate] • Target: Dot Product and Orthogonality

A line has vector equation $\mathbf{r} = (0; -3; 5) + t(3; -4; -3)$. Find the point when $t = 3$.

Final Answer: $(9, -15, -4)$. Working through this vectors step:

$\mathbf{r} = (0; -3; 5) + (3)(3; -4; -3) = (9, -15, -4)$

10 [Intermediate] • Target: The Vector Cross Product Formula

A line has vector equation $\mathbf{r} = (6; -6; -6) + t(1; -1; -2)$. Find the point when $t = 0$.

Final Answer: $(6, -6, -6)$. Working through this vectors step:

$\mathbf{r} = (6; -6; -6) + (0)(1; -1; -2) = (6, -6, -6)$

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11 [Intermediate] • Target: Area of Triangles and Parallelograms

A line has vector equation $\mathbf{r} = (4;\ -4;\ -6) + t(2;\ -3;\ -3)$. Find the point when $t = -2$.

Final Answer: $(0, 2, 0)$. Working through this vectors step:

$\mathbf{r} = (4;\ -4;\ -6) + (-2)(2;\ -3;\ -3) = (0, 2, 0)$

12 [Intermediate] • Target: Scalar Triple Product & Volume

A line has vector equation $\mathbf{r} = (-2; 1;\ -2) + t(-1; 4; 4)$. Find the point when $t = 1$.

Final Answer: $(-3, 5, 2)$. Working through this vectors step:

$\mathbf{r} = (-2; 1;\ -2) + (1)(-1; 4; 4) = (-3, 5, 2)$

13 [Intermediate] • Target: Dot Product and Orthogonality

A line has vector equation $\mathbf{r} = (4; 3; 3) + t(2; 2;\ -1)$. Find the point when $t = 3$.

Final Answer: $(10, 9, 0)$. Working through this vectors step: $\mathbf{r} = (4; 3; 3) + (3)(2; 2;\ -1) = (10, 9, 0)$

14 [Intermediate] • Target: The Vector Cross Product Formula

Use the compound angle formula for cosine to show that $\cos(180^\circ - x) = -\cos x,$ then find $\cos 150^\circ$. .

Final Answer: $-3/2$. $\cos(180^\circ - x) = \cos 180^\circ \cos x + \sin 180^\circ \sin x = -\cos x.$ So $150^\circ = (180^\circ - 30^\circ) = -30^\circ = -3/2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Area of Triangles and Parallelograms • Cross-Lesson: 1.04 Partial Fractions

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-5;\ -2; 2)$ passing through $P(-3,\ -4,\ -3)$.

Final Answer: $-5x - 2y + 2z = 17$. Working through this vectors step:

$d = \mathbf{n} \cdot P = (-5)(-3) + (-2)(-4) + (2)(-3) = 17$

16 [Challenging] • Target: Scalar Triple Product & Volume • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-6;\ -6;\ -3)$ passing through $P(2, 6,\ -2)$.

Final Answer: $-6x - 6y - 3z = -42$. Working through this vectors step:

$d = \mathbf{n} \cdot P = (-6)(2) + (-6)(6) + (-3)(-2) = -42$

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17 [Challenging] • Target: Dot Product and Orthogonality • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-1; 3; 4)$ passing through $P(-5, -6, -1)$.

Final Answer: $-x + 3y + 4z = -17$. Working through this vectors step:

$d = \mathbf{n} \cdot P = (-1)(-5) + (3)(-6) + (4)(-1) = -17$

18 [Challenging] • Target: The Vector Cross Product Formula • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (1; -3; 4)$ passing through $P(-6, 6, -1)$.

Final Answer: $x - 3y + 4z = -28$. Working through this vectors step:

$d = \mathbf{n} \cdot P = (1)(-6) + (-3)(6) + (4)(-1) = -28$

19 [Challenging] • Target: Area of Triangles and Parallelograms • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-4; -6; 4)$ passing through $P(0, 4, -2)$.

Final Answer: $-4x - 6y + 4z = -32$. Working through this vectors step:

$d = \mathbf{n} \cdot P = (-4)(0) + (-6)(4) + (4)(-2) = -32$

20 [Challenging] • Target: Scalar Triple Product & Volume • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (-3; 5; -5)$ passing through $P(-5, 5, -3)$.

Final Answer: $-3x + 5y - 5z = 55$. Working through this vectors step:

$d = \mathbf{n} \cdot P = (-3)(-5) + (5)(5) + (-5)(-3) = 55$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Dot Product and Orthogonality

Calculate the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = [2, -3, 5]$ and $\mathbf{v} = [4, 1, -2]$.

a. -5 ✓

b. 5

c. 15

d. -1

$\mathbf{u} \cdot \mathbf{v} = (2)(4) + (-3)(1) + (5)(-2) = 8 - 3 - 10 = -5$.

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22 [Foundational] • Target: Dot Product and Orthogonality

For what value of k are the vectors $\mathbf{a} = [k, 3, -2]$ and $\mathbf{b} = [4, -2, 5]$ orthogonal (perpendicular)?

a. 4 ✓

b. 2

c. -4

d. 5

Orthogonal vectors satisfy $\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 4k + 3(-2) + (-2)(5) = 0 \Rightarrow 4k - 6 - 10 = 0 \Rightarrow 4k = 16 \Rightarrow k = 4$.

23 [Foundational] • Target: Dot Product and Orthogonality

Find the angle θ between $\mathbf{u} = [1, 1, 0]$ and $\mathbf{v} = [0, 1, 1]$.

a. $\frac{\pi}{3}$ **(**60°**)** ✓

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{6}$ (30°)

d. $\frac{\pi}{2}$ (90°)

$\mathbf{u} \cdot \mathbf{v} = 1(0) + 1(1) + 0(1) = 1$. $|\mathbf{u}| = \sqrt{2}$, $|\mathbf{v}| = \sqrt{2}$. $\cos\theta = \frac{1}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$.

24 [Foundational] • Target: The Vector Cross Product Formula

Compute the cross product $\mathbf{u} \times \mathbf{v}$ where $\mathbf{u} = [1, 2, 3]$ and $\mathbf{v} = [4, 5, 6]$.

a. $[-3, 6, -3]$ ✓

b. $[3, -6, 3]$

c. $[-3, -6, -3]$

d. $[7, -2, -3]$

$\mathbf{u} \times \mathbf{v} = [2(6) - 3(5), -(1(6) - 3(4)), 1(5) - 2(4)] = [12 - 15, -(6 - 12), 5 - 8] = [-3, 6, -3]$.

25 [Foundational] • Target: The Vector Cross Product Formula

What is the geometric relationship between the cross product vector $\mathbf{u} \times \mathbf{v}$ and the vectors $\mathbf{u}$ and $\mathbf{v}$?

a. **$\mathbf{u} \times \mathbf{v}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$** ✓

b. $\mathbf{u} \times \mathbf{v}$ is parallel to $\mathbf{u} + \mathbf{v}$

c. $\mathbf{u} \times \mathbf{v}$ bisects the angle between $\mathbf{u}$ and $\mathbf{v}$

d. $\mathbf{u} \times \mathbf{v}$ lies in the plane spanned by $\mathbf{u}$ and $\mathbf{v}$

By definition, the cross product produces a vector that is orthogonal to both input vectors, directed by the right-hand rule.

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26 [Foundational] • Target: The Vector Cross Product Formula

If $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ for non-zero vectors $\mathbf{u}$ and $\mathbf{v}$, what does this indicate?

a. The vectors are parallel (collinear) ✓

b. The vectors are perpendicular

c. The vectors have unit magnitude

d. The vectors are identical

$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin\theta = 0 \Rightarrow \sin\theta = 0 \Rightarrow \theta = 0$ or π, meaning the vectors are parallel.

27 [Foundational] • Target: Area of Triangles and Parallelograms

Find the area of the parallelogram spanned by vectors $\mathbf{a} = [2, 0, 1]$ and $\mathbf{b} = [1, 3, -2]$.

a. $\sqrt{59}$ ✓

b. $\sqrt{35}$

c. 14

d. $\frac{\sqrt{59}}{2}$

$\mathbf{a} \times \mathbf{b} = [0(-2) - 1(3), -(2(-2) - 1(1)), 2(3) - 0(1)] = [-3, 5, 6]$. Area $= |\mathbf{a} \times \mathbf{b}| = \sqrt{(-3)^2 + 5^2 + 6^2} = \sqrt{9 + 25 + 36} = \sqrt{70}$.

28 [Foundational] • Target: Area of Triangles and Parallelograms

Find the area of the triangle with vertices $A(1, 0, 0), B(0, 2, 0), C(0, 0, 3)$.

a. $\frac{7}{2}$ ✓

b. 7

c. $\sqrt{14}$

d. 3

$\vec{AB} = [-1, 2, 0]$ and $\vec{AC} = [-1, 0, 3]$. $\vec{AB} \times \vec{AC} = [6, 3, 2]$. Magnitude $= \sqrt{36 + 9 + 4} = \sqrt{49} = 7$. Triangle area $= \frac{1}{2}(7) = 3.5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Area of Triangles and Parallelograms

If two vectors have lengths $|\mathbf{u}| = 6$ and $|\mathbf{v}| = 5$, and the angle between them is 30°, what is $|\mathbf{u} \times \mathbf{v}|$?

a. 15 ✓

b. $15\sqrt{3}$

c. 30

d. $30\sqrt{3}$

$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin(30^\circ) = (6)(5)\left(\frac{1}{2}\right) = 15.$

30 [Intermediate] • Target: Scalar Triple Product & Volume

What geometric quantity is given by the absolute value of the scalar triple product $|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$?

a. **The volume of the parallelepiped formed by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$** ✓

b. The surface area of the prism formed by $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$

c. The length of the main diagonal vector

d. The perimeter of the base

The scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ represents the volume of the parallelepiped with adjacent edges $\mathbf{a}, \mathbf{b}, \mathbf{c}$.

31 [Intermediate] • Target: Scalar Triple Product & Volume

Calculate the volume of the parallelepiped with edges $\mathbf{a} = [1, 0, 0]$, $\mathbf{b} = [0, 2, 0]$, $\mathbf{c} = [1, 1, 3]$.

a. 6 ✓

b. 3

c. 12

d. 2

Volume $= |\det[[1, 0, 0], [0, 2, 0], [1, 1, 3]]| = |1(2 \times 3 - 0)| = 6.$

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32 [Intermediate] • Target: Scalar Triple Product & Volume

If $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, what does this reveal about the three vectors?

a. The three vectors are coplanar (lie in the same plane) ✓

b. The three vectors are mutually perpendicular

c. At least one vector must be the zero vector

d. The vectors form an equilateral triangle

A zero scalar triple product implies zero enclosed volume, meaning the three vectors lie in a single common geometric plane (coplanar).

33 [Intermediate] • Target: Dot Product and Orthogonality

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (5; -5; 5)$ passing through $P(-6, 2, 2)$.

a. $5x - 5y + 5z = -28$

b. $5x - 5y + 5z = -27$

c. $5x - 5y + 5z = -29$

d. $5x - 5y + 5z = -30$ ✓

Working through this vectors step: $d = \mathbf{n} \cdot P = (5)(-6) + (-5)(2) + (5)(2) = -30$

34 [Intermediate] • Target: The Vector Cross Product Formula

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (6; 5; -2)$ passing through $P(-1, -1, -6)$.

a. $6x + 5y - 2z = -2$

b. $6x + 5y - 2z = 0$

c. $6x + 5y - 2z = -1$

d. $6x + 5y - 2z = 1$ ✓

Working through this vectors step: $d = \mathbf{n} \cdot P = (6)(-1) + (5)(-1) + (-2)(-6) = 1$

35 [Intermediate] • Target: Area of Triangles and Parallelograms

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (4; -2; 6)$ passing through $P(-3, 6, 1)$.

a. $4x - 2y + 6z = -18$ ✓

b. $4x - 2y + 6z = -21$

c. $4x - 2y + 6z = -19$

d. $4x - 2y + 6z = -15$

Working through this vectors step: $d = \mathbf{n} \cdot P = (4)(-3) + (-2)(6) + (6)(1) = -18$

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36 [Intermediate] • Target: Scalar Triple Product & Volume

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (3;\ -6;\ -6)$ passing through $P(1, 0, -1)$.

a. $3x - 6y - 6z = 11$

b. $3x - 6y - 6z = 9$ ✓

c. $3x - 6y - 6z = 6$

d. $3x - 6y - 6z = 8$

Working through this vectors step: $d = \mathbf{n} \cdot P = (3)(1) + (-6)(0) + (-6)(-1) = 9$

37 [Intermediate] • Target: Dot Product and Orthogonality

Find the equation $ax + by + cz = d$ of the plane with normal vector $\mathbf{n} = (3;\ 2;\ -1)$ passing through $P(6, 2, -5)$.

a. $3x + 2y - z = 25$

b. $3x + 2y - z = 28$

c. $3x + 2y - z = 27$ ✓

d. $3x + 2y - z = 30$

Working through this vectors step: $d = \mathbf{n} \cdot P = (3)(6) + (2)(2) + (-1)(-5) = 27$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Vector Cross Product Formula • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Using the compound angle formula, expand $\sin(x + 90^\circ)$.

a. $\cos x$ ✓

b. $-\cos x$

c. $\sin x$

d. $-\sin x$

$\sin(x + 90^\circ) = \sin x \cos 90^\circ + \cos x \sin 90^\circ = 0 + \cos x = \cos x$.

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39 [Challenging] • Target: Area of Triangles and Parallelograms • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

The diagram shows two similar triangles ABC and ADE, where D lies on AB and E lies on AC. It is given that AD = 6 cm, DB = 4 cm, and DE = 12 cm. Find the length of BC.

a. 18 cm

b. **20 cm** ✓

c. 16 cm

d. 24 cm

Since triangles ABC and ADE are similar, the ratio of corresponding sides is equal. The ratio of similarity between triangle ADE and triangle ABC can be found by comparing the lengths of corresponding sides AD and AB. We have AB = AD + DB = 6 + 4 = 10 cm. The ratio of similarity, k, is $\frac{AB}{AD} = \frac{10}{6} = \frac{5}{3}$. Alternatively, we can consider the ratio of similarity from ADE to ABC as $\frac{AD}{AB} = \frac{6}{10} = \frac{3}{5}$. Using this ratio, we can find the length of BC. The corresponding s.

40 [Challenging] • Target: Scalar Triple Product & Volume • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Consider the points $A(1,\ -2, 3)$, $B(4, 0, 5)$, and $C(p, 1, 2)$. If the vectors $\vec{AB}$ and $\vec{AC}$ are perpendicular, find the value of p.

a. $p = -5$

b. $p = -4$

c. $p = 4$ ✓

d. $p = 5$

First, find the vector $\vec{AB}$: $\vec{AB} = B - A = (4 - 1, 0 - (-2), 5 - 3) = (3, 2, 2)$. Next, find the vector $\vec{AC}$: $\vec{AC} = C - A = (p - 1, 1 - (-2), 2 - 3) = (p - 1, 3, -1)$. Two vectors are perpendicular if their dot product is zero. Therefore, $\vec{AB} \cdot \vec{AC} = 0$. $(3)(p - 1) + (2)(3) + (2)(-1) = 0$ $3p - 3 + 6 - 2 = 0$ $3p + 1 = 0$ $3p = -1$ $p = -\frac{1}{3}$

41 [Challenging] • Target: Dot Product and Orthogonality • Cross-Lesson: 4.02 The Poisson Distribution

Find d in the equation $ax + by + cz = d$ of the plane through $A(-1, 3, -1)$, $B(-2, 0, 3)$, and $C(-4, 3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. **3** ✓

b. 2

c. 8

d. 7

Working through this vectors step: $\vec{AB} = (-1;\ -3; 4)$, $\vec{AC} = (-3; 0; 5)$, $\mathbf{n} = (-15;\ -7;\ -9)$, $d = \mathbf{n} \cdot A = 3$

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42 [Challenging] • Target: The Vector Cross Product Formula • Cross-Lesson: 5.01 Further Integration Techniques

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, 1, -4)$, $B(1, 4, -2)$, and $C(-1, -3, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 6

b. 9

c. 7

d. **14** ✓

Working through this vectors step: $\vec{AB} = (-1; 3; 2)$, $\vec{AC} = (-3; -4; 8)$, $\mathbf{n} = (32; 2; 13)$, $d = \mathbf{n} \cdot A = 14$

43 [Challenging] • Target: Area of Triangles and Parallelograms • Cross-Lesson: 5.02 Differential Equations

Find d in the equation $ax + by + cz = d$ of the plane through $A(-1, -3, 1)$, $B(-3, 4, 2)$, and $C(-4, -2, 4)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. -7

b. -17

c. **-10** ✓

d. -13

Working through this vectors step: $\vec{AB} = (-2; 7; 1)$, $\vec{AC} = (-3; 1; 3)$, $\mathbf{n} = (20; 3; 19)$, $d = \mathbf{n} \cdot A = -10$

44 [Challenging] • Target: Scalar Triple Product & Volume • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Find d in the equation $ax + by + cz = d$ of the plane through $A(-1, -1, -4)$, $B(-4, 4, 4)$, and $C(1, 2, 0)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 51

b. 39

c. **52** ✓

d. 37

Working through this vectors step: $\vec{AB} = (-3; 5; 8)$, $\vec{AC} = (2; 3; 4)$, $\mathbf{n} = (-4; 28; -19)$, $d = \mathbf{n} \cdot A = 52$

45 [Challenging] • Target: Dot Product and Orthogonality • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Find d in the equation $ax + by + cz = d$ of the plane through $A(2, -3, 4)$, $B(2, -1, 2)$, and $C(-3, 2, 1)$, using normal $\mathbf{n} = \vec{AB} \times \vec{AC}$.

a. 4

b. 24

c. 20

d. **18** ✓

Working through this vectors step: $\vec{AB} = (0; 2; -2)$, $\vec{AC} = (-5; 5; -3)$, $\mathbf{n} = (4; 10; 10)$, $d = \mathbf{n} \cdot A = 18$

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